Physical Chemistry of Materials

Lecture 09

1차 중간고사 안내 및 복습

1차 중간고사 안내

- 시간: 10월 13일(화) 오후 1:30~2:30
- 장소: 화학관 414호
- 신분증 지참(모바일 학생증도 가능)
- 계산기 불필요

시험 범위

기초 열역학(2강)

상 전이(3강)

통계역학의 원리(4강)

정준 앙상블의 응용(5강)

이상 기체/화학 평형(6강)

상호작용하는 입자들(7강)

대정준 앙상블(8강)

시험 문제

총 여섯 문제

개념을 묻는 문제: 두 문제

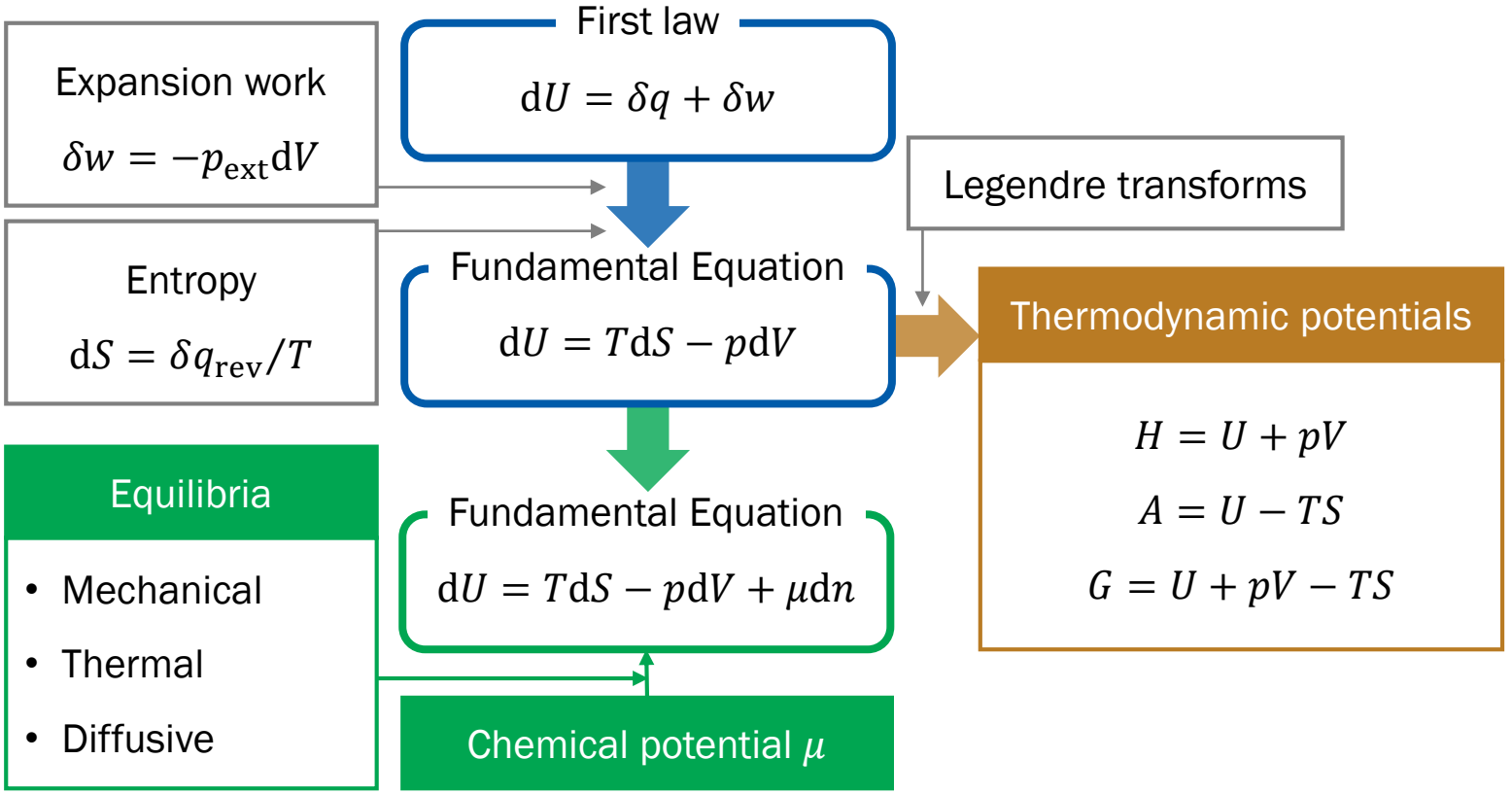
공식을 유도하는 문제: 세 문제

응용 문제: 한 문제

소재물리화학의 구성

고전 열역학	고체	연성물질
열역학 복습	고체의 구조	액체
상 전이	고체의 진동	유리
통계역학	고체의 전기적 성질	고분자의 구조
기초 이론	고체의 자기적 성질	고분자 용액
이상 기체	표면화학	고분자의 점탄성
상호작용 계	반도체, 나노화학, …	생체 고분자

Basic Thermodynamics



Phase Transitions

- Stability of phases: stable, metastable, unstable
- Binodal and spinodal
- Example: phase separation
  - Phase diagram
  - Kinetics: spinodal decomposition vs. nucleation and growth
- Nucleation and growth models
  - Classical nucleation theory
  - Ostwald ripening

Statistical Mechanics

Ensemble

↓

Principle of maximum entropy

$$S = -k_{\mathrm{B}} \sum_i p_i \ln p_i$$

↓

Constrained optimization

↓

Determination of multipliers $(\lambda, \beta, \alpha, \cdots)$

Ensembles

	Microcanonical	Canonical	Grand canonical
Fluctuation	No	E	E and N
p_i	$1/W$	$e^{-\beta E_i}/Z$	$e^{-\beta E_i - \alpha N_i}/\mathcal{Z}$
Normalization constant	Number of microstates $W = \sum_i 1$	Partition function $Z = \sum_i e^{-\beta E_i}$	Grand partition function $\mathcal{Z} = \sum_i e^{-\beta E_i - \alpha N_i}$
Multipliers		$\beta = 1/k_\mathrm{B}T$	$\beta = 1/k_\mathrm{B}T$ $\alpha = -\tilde{\mu}/k_\mathrm{B}T$
Thermodynamic potential	Entropy $S = k_\mathrm{B} \ln W$	Helmholtz energy $A = -k_\mathrm{B}T \ln Z$	Grand potential $\phi = -k_\mathrm{B}T \ln \mathcal{Z}$

Useful Trick

For $p_i = C_i e^{-kx_i}/Z$ and $Z = \sum_i C_i e^{-kx_i}$,

$$\langle x \rangle = \sum_i x_i p_i = \frac{1}{Z} \sum_i x_i C_i e^{-kx_i}$$

but

$$\frac{\partial \ln Z}{\partial k} = \frac{1}{Z}\frac{\partial Z}{\partial k} = \frac{1}{Z}\frac{\partial}{\partial k}\sum_i C_i e^{-kx_i} = -\frac{1}{Z}\sum_i x_i C_i e^{-kx_i}$$

Hence,

$$\langle x \rangle = -\frac{\partial \ln Z}{\partial k}$$

1-Particle Partition Functions

- Discrete microstates

$$z = \sum_{s} e^{-\beta E(s)}$$

- Classical particles (in 1D)

$$z = \frac{1}{h} \iint e^{-\beta E(x,p)} \, \mathrm{d}x \mathrm{d}p$$

N-Particle Partition Functions

- Independent & distinguishable particles

$$Z = z^N \text{ (z: one-particle partition function)}$$

- Independent & indistinguishable particles

$$Z = \frac{1}{N!} z^N$$

- Interacting particles: depends on interactions

Applications: Canonical Ensemble

- Ideal gas: QM vs. CM
  - Translation – particle in a box
  - Vibration – simple harmonic oscillator
  - Rotation – rigid rotor
- Equilibrium constant of ideal gas
- Real gas
  - Intramolecular interactions
  - Evaluation of partition function

Grand Partition Functions

$$\mathcal{Z} = \sum_i e^{-\alpha N_i - \beta E_i}$$

$$\mathcal{Z} = \sum_N e^{-\alpha N} \left(\sum_s e^{-\beta E_s(N)} \right)$$

$$\mathcal{Z} = \sum_N \left(e^{\mu/RT} \right)^N Z(N)$$

Calculation of $\mathcal{Z}$

$$\mathcal{Z} = \sum_N \left(e^{\mu/RT}\right)^N Z(N)$$

- Independent & distinguishable particles

$$\mathcal{Z} = \sum_N \left(e^{\mu/RT}\right)^N z^N = \sum_N \left(ze^{\mu/RT}\right)^N$$

- Independent & indistinguishable particles

$$\mathcal{Z} = \sum_N \left(e^{\mu/RT}\right)^N \frac{z^N}{N!} = \sum_N \frac{1}{N!}\left(ze^{\mu/RT}\right)^N = \exp\left(ze^{\mu/RT}\right)$$

Average Occupation Number

For a specific microstate with energy ϵ,

- Classical particles (Maxwell-Boltzmann distribution)

$$N_{\mathrm{MB}}(\epsilon) = e^{-(\epsilon-\tilde{\mu})/k_{\mathrm{B}}T}$$

- Fermions (Fermi-Dirac distribution)

$$N_{\mathrm{FD}}(\epsilon) = \frac{1}{e^{(\epsilon-\tilde{\mu})/k_{\mathrm{B}}T} + 1}$$

- Bosons (Bose-Einstein distribution)

$$N_{\mathrm{BE}}(\epsilon) = \frac{1}{e^{(\epsilon-\tilde{\mu})/k_{\mathrm{B}}T} - 1}$$